1 Method Description

1.1 Derivation of Main Equations

By performing a force balance (so assuming no acceleration) on the 2D body we get that

$$-\sigma_{xx}A_x + \left(\sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x}\Delta x\right)A_x - \sigma_{yx}A_y + \left(\sigma_{yx} + \frac{\partial \sigma_{yx}}{\partial y}\Delta y\right)A_y + S_x = 0 \tag{1}$$

and

$$-\sigma_{yy}A_y + \left(\sigma_{yy} + \frac{\partial \sigma_{yy}}{\partial y}\Delta y\right)A_y - \sigma_{xy}A_x + \left(\sigma_{xy} + \frac{\partial \sigma_{xy}}{\partial x}\Delta x\right)A_x + S_y = 0. \tag{2}$$

Note that for example A_x is the area of the face in the y-z plane. Its normal is in the x-direction. Here, S is the internal force. By canceling out some like terms we have

$$\frac{\partial \sigma_{xx}}{\partial x}\Delta x A_x + \frac{\partial \sigma_{yx}}{\partial y}\Delta y A_y + S_x = 0 \tag{3}$$

and

$$\frac{\partial \sigma_{yy}}{\partial y}\Delta y A_y + \frac{\partial \sigma_{xy}}{\partial x}\Delta x A_x + S_y = 0. \tag{4}$$

Next, if we assume a rectangular control volume of dimensions $\Delta x \times \Delta y \times \Delta z$, $\Delta y A_y$ and $\Delta x A_x$ are both equivalent to V. S then becomes f, which represents the force per unit volume in that direction. Also, the shear stresses can be written with τ.

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + f_x = 0 \tag{5}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y = 0, \tag{6}$$

or

$$\nabla \cdot \boldsymbol{\sigma} = -\boldsymbol{f}, \tag{7}$$

where

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix}. \tag{8}$$

It has been assumed that $\tau_{xy} = \tau_{yx}$ due to an angular moment balance. These equations, when written in matrix form are

$$\begin{bmatrix} \frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = - \begin{bmatrix} f_x \\ f_y \end{bmatrix} \tag{9}$$

or simply

$$B^T \boldsymbol{\sigma} = -\boldsymbol{f}, \tag{10}$$

where B is a differential operator that relates stresses to force equilibrium. It is assumed that displacements are sufficiently small that equilibrium conditions that were established for the initial geometry remain valid. Note that the stresses here are total stresses. In the case where we have pore pressure, we replace total stress with the definition of effective stress, $\boldsymbol{\sigma} = \boldsymbol{\sigma'} + \alpha p\mathbf{I}$, with $\mathbf{I}$ being the identity matrix and α the Biot coefficient, such that our equation becomes

$$\nabla \cdot \left(\boldsymbol{\sigma'} + \alpha p\mathbf{I}\right) = -\boldsymbol{f}. \tag{11}$$

Here, compression is taken as positive. This results in

$$\nabla \cdot \boldsymbol{\sigma'} + \nabla\left(\alpha p\right) = -\boldsymbol{f}, \tag{12}$$

or

$$B^T \boldsymbol{\sigma'} + \nabla\left(\alpha p\right) = -\boldsymbol{f}, \tag{13}$$

where

$$\nabla\left(\alpha p\right) = \alpha \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \end{bmatrix}, \tag{14}$$

for a spatially-constant α.

1.2 Hooke's Law

Hooke's Law reads,

$$\sigma_{ij}^{'} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}, \tag{15}$$

where $\sigma^{'}$ is an effective stress. $\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$, so

$$\sigma_{ij}^{'} = \frac{E\nu}{\left(1+\nu\right)\left(1-2\nu\right)} \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}, \tag{16}$$

and $\mu = \frac{E}{2(1+\nu)}$, so

$$\sigma_{ij}^{'} = \frac{E\nu}{\left(1+\nu\right)\left(1-2\nu\right)} \epsilon_{kk} \delta_{ij} + \frac{E}{\left(1+\nu\right)} \epsilon_{ij}. \tag{17}$$

Since $\epsilon_{kk} = \epsilon_{xx} + \epsilon_{yy}$ in Cartesian coordinates for plane strain in x-y plane ($\epsilon_{zz} = 0$),

$$\sigma'_{xx} = \frac{E}{(1+\nu)(1-2\nu)} \left((1-\nu)\,\epsilon_{xx} + \nu\epsilon_{yy} \right), \tag{18}$$

$$\sigma'_{yy} = \frac{E}{(1+\nu)(1-2\nu)} \left((1-\nu)\,\epsilon_{yy} + \nu\epsilon_{xx} \right), \tag{19}$$

and

$$\tau_{xy} = \frac{E}{2(1+\nu)}\gamma_{xy}. \tag{20}$$

Note that the shear strain here is the engineering shear strain, such that $\gamma_{xy} = 2\epsilon_{xy}$. These equations can be written in matrix form as:

$$\begin{bmatrix} \sigma'_{xx} \\ \sigma'_{yy} \\ \tau_{xy} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2v}{2} \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}, \tag{21}$$

which can also be written as

$$\boldsymbol{\sigma}' = D\boldsymbol{\epsilon}, \tag{22}$$

where

$$\boldsymbol{\sigma}' = \begin{bmatrix} \sigma'_{xx} \\ \sigma'_{yy} \\ \tau_{xy} \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}. \tag{23}$$

Although not solved for explicitly, the out-of-plane effective stress is given by $\sigma'_{zz} = \frac{E\nu}{(1+\nu)(1-2\nu)}(\epsilon_{xx} + \epsilon_{yy})$.

1.3 Rewriting the Strains

The definition for ϵ under the assumption of small strains is

$$\epsilon = \frac{\Delta L}{L}. \tag{24}$$

The limit as the length of our system goes to zero causes

$$\lim_{L \to 0} \frac{\Delta L}{L} = \frac{du}{dx}. \tag{25}$$

This can be done for both directions such that

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} \tag{26}$$

and

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}. \tag{27}$$

The engineering shear strain, $\gamma_{ij} = 2\epsilon_{ij}$, for small displacement gradients is given by

$$\gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}. \tag{28}$$

These equations can then be written in matrix form yielding

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix} \tag{29}$$

or simply

$$\widetilde{\epsilon} = B\widetilde{u} \tag{30}$$

and also

$$\epsilon = Bu \tag{31}$$

Terms with a tilde are continuous whereas when displacement and strain are written without a tilde it means that they are nodal values.

Setting Up The System Of Equations

By combining Equations ??, ??, and ?? we get

$$B^T DBu = -f - \alpha\nabla p. \tag{32}$$

Basis Functions

The displacement, $\widetilde{u}$, can be found at any point in the element through the use of the values at the nodes and basis functions such that

$$\widetilde{u} = N_1u_1 + N_2u_2 + N_3u_3 + N_4u_4 \tag{33}$$

for an element with four points or, more generally in matrix form,

$$\widetilde{u} = Nu. \tag{34}$$

Here, N represents the basis functions and u represents the value of displacement (in both x and y) at the nodes. For an element with four nodes, this looks like

$$\widetilde{u} = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix} \begin{bmatrix} u_{x1} \\ u_{y1} \\ u_{x2} \\ u_{y2} \\ u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix} \tag{35}$$

For a square element with linear basis functions with the node numbering beginning in the upper left and moving horizontally before vertically, the basis functions are

$$N_1 = \left(1 - \frac{x}{a}\right)\left(1 - \frac{y}{b}\right) \tag{36}$$

$$N_2 = \frac{x}{a}\left(1 - \frac{y}{b}\right) \tag{37}$$

$$N_3 = \left(1 - \frac{x}{a}\right)\frac{y}{b} \tag{38}$$

$$N_4 = \frac{x}{a}\frac{y}{b} \tag{39}$$

Here, a is the length of the element in the x direction and b is the length of the element in the y direction. Now, Equation ?? which we had previously is written like

$$B^T DBNu = -f - \nabla p \tag{40}$$

1.4 Including Body Forces and Pore Pressures

Multiplying the governing equilibrium equation by the test functions and integrating over the element domain yields

$$\int_{\Omega_e} N^T B^T DBNu \, d\Omega = -\int_{\Omega_e} N^T f \, d\Omega - \int_{\Omega_e} B^T(\alpha p) \, d\Omega. \tag{41}$$

The left-hand side defines the element stiffness matrix, while the right-hand side consists of equivalent nodal forces due to body forces and pore pressure.

$$K_e u_e = F_e + P_e. \tag{42}$$

Note that practically, the force due to pore pressure, P_{Ax}, on node A in the x-direction would then be give by

$$P_{Ax} = \alpha \frac{dy}{2} \left(P_1 + P_3 - P_2 - P_4\right) \tag{43}$$

The same procedure is used to construct equivalent nodal forces arising from distributed loads and body forces.

1.5 Assembly of the Global System

For each finite element, the weak form leads to the element-level system

$$\boldsymbol{K}_e \boldsymbol{u}_e = \boldsymbol{F}_e + \boldsymbol{P}_e, \tag{44}$$

where the element stiffness matrix is defined as

$$\boldsymbol{K}_e = \int_{\Omega_e} \boldsymbol{B}_e^T D \boldsymbol{B}_e \, d\Omega, \tag{45}$$

where $\boldsymbol{B}_e = BN$.

For a quadrilateral element with four nodes, this results in an 8×8 matrix.

The constitutive matrix for plane strain is given by

$$D = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}. \tag{46}$$

Additionally,

$$\boldsymbol{B}_e = BN = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix}. \tag{47}$$

Now, for example, $\boldsymbol{K}_e\,(1,1)$ can be found as:

$$\boldsymbol{K}_e\,(1,1) = \frac{E}{(1+\nu)\,(1-2\nu)} \int_0^a \int_0^b \left[(1-\nu) \left(\frac{\partial N_1}{\partial x} \right)^2 + \frac{1-2\nu}{2} \left(\frac{\partial N_1}{\partial y} \right)^2 \right] \partial\xi \partial\eta \tag{48}$$

where the basis functions can be directly plugged in and integrated. In practice, element integrals are typically evaluated using Gauss quadrature, particularly for distorted quadrilateral elements where analytical integration becomes cumbersome. It is possible to take each derivative manually and fill up the local (and ultimately the global) stiffness matrix in this way. The element matrices and force vectors are assembled into a global system according to the mesh connectivity. After application of the displacement boundary conditions, the resulting linear system is

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F} + \boldsymbol{P}, \tag{49}$$

where $\boldsymbol{K}$ is the global stiffness matrix, $\boldsymbol{u}$ is the vector of nodal displacements, and $\boldsymbol{F}$ and $\boldsymbol{P}$ are the assembled body-force and pore-pressure load vectors, respectively. The nodal displacements are then obtained by solving this linear system.

1.6 Strain recovery

Once the nodal displacements have been obtained, strains are evaluated within each element from the strain-displacement relationship,

$$\boldsymbol{\epsilon} = \boldsymbol{B}_e \boldsymbol{u}_e. \tag{50}$$

1.7 Stress recovery

The stresses can then simply be solved for in the following manner:

$$\boldsymbol{\sigma}' = D\boldsymbol{\epsilon} \tag{51}$$

and total stress can be found with

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' + \alpha p \mathbf{I}. \tag{52}$$

Under the plane-strain assumption, the out-of-plane stress component may also be recovered from

$$\sigma'_{zz} = \frac{E\nu}{(1+\nu)(1-2\nu)}(\epsilon_{xx} + \epsilon_{yy}). \tag{53}$$

1.8 Volumetric strain

The volumetric strain is obtained as the trace of the strain tensor and is used to couple deformation to the fluid flow model through porosity evolution.

$$\epsilon_v = \mathbf{m}^T \mathbf{B}_e \mathbf{u}_e, \tag{54}$$

where

$$\mathbf{m} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \tag{55}$$

2 Coupling

The volumetric strain can then be used to update porosity in the fluid flow model as,

$$\phi = \phi_0 + (1 - \phi_0)\,\epsilon_v, \tag{56}$$

where ϕ is porosity and ϕ_0 is a reference porosity. The fluid flow model then iterates until convergence, returning pore pressure. These pore pressure are input into the mechanics model to find volumetric strain. This volumetric strain is passed back to the fluid flow model and the sequential coupling repeats until global convergence of both models. At that point, the simulator advances to the next time step.